

Tutorial 1 : Boolean algebra (Part 1)

Exercise 1:

Establish the truth tables for the following functions

- $F = A.B + B.C + A.C$
- $G = \bar{X} + Y.Z + \bar{Y}.Z.T$
- $H = (A+B) . (A + B + C)$
- $I = (X + \bar{Z}) . (X + \bar{T} + Z) . (Y . Z)$

Exercise 2:

We respectively denote the operators 'XOR', 'NAND', and 'NOR' as: $\oplus$, $\uparrow$ et $\downarrow$

Show using truth tables that:

- $A \oplus B = \bar{A}.B + A.\bar{B}$
- $A \uparrow B \uparrow C = \overline{A.B.C}$
- $A \downarrow B \downarrow C \downarrow D = \overline{A + B + C + D}$

Exercise 3:

Prove (without using a truth table) that:

- $A + (A \cdot B) = A$
- $A \cdot (A + B) = A$
- $A \cdot (\bar{A} + B) = A \cdot B$
- $A + (\bar{A} \cdot B) = A + B$

Exercise 4:

Prove using the truth table the Morgan's Theorem:

- $\overline{A + B} = \bar{A} \cdot \bar{B}$
- $\overline{A \cdot B} = \bar{A} + \bar{B}$